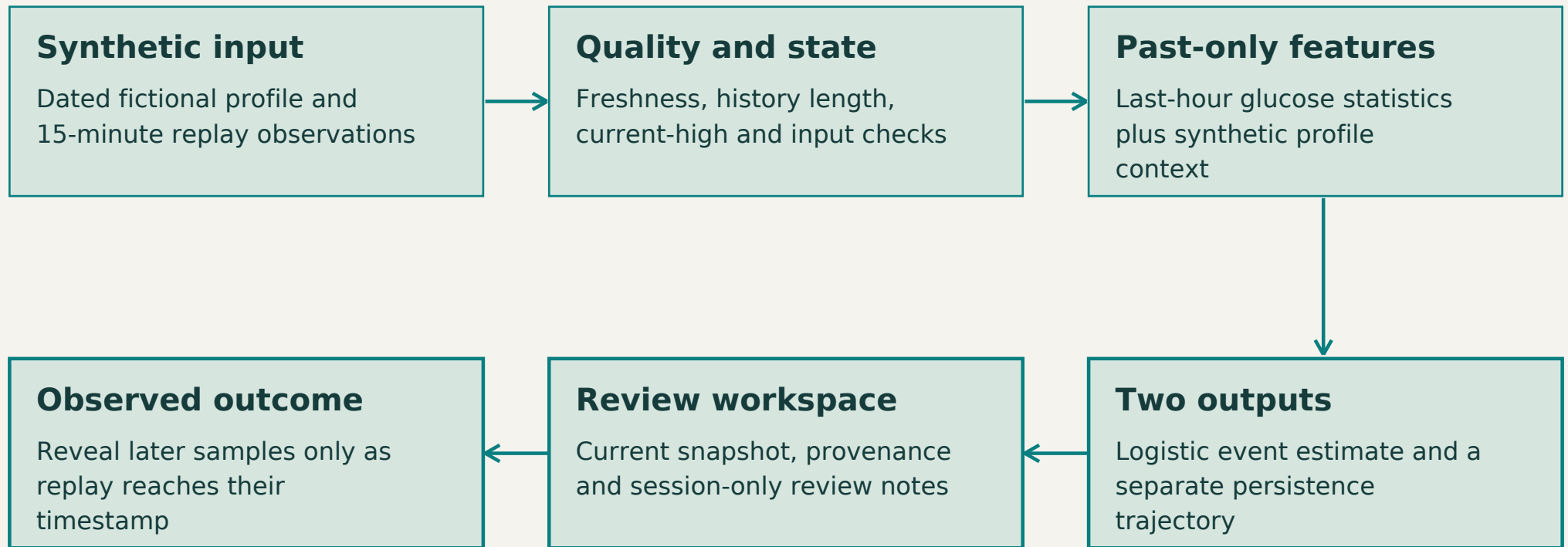


GlucoTwin architecture

Implemented browser runtime and separate offline evidence pipelines



React + TypeScript + Vite. Synthetic coefficients, scaling values, fixtures and evidence ship as JSON. Inference runs in the browser without a Python service, API key, login, or real-patient upload.

Sources: src/App.tsx, src/lib/engine.ts, public/model.json, public/model-demo.json. The prototype has no physical sensor/EHR connection or autonomous treatment loop.

Offline model and evidence pipelines

The public browser never runs the real-data benchmark model

Synthetic engineering pipeline

model/train.py creates 240 fictional people. A fixed patient split assigns 144 to training, 48 to validation and 48 to test. StandardScaler and logistic regression fit training people only. Validation selects thresholds and residual bands. Exported JSON supports browser parity tests and selected synthetic replay cases.

Separate retrospective pipeline

model/benchmark_real.py reads ShanghaiT2DM locally from outside the repository. All 109 sessions stay grouped within 100 people. The 60/20/20 patient split supports separate static, dynamic and fused models. Only aggregate results leave this pipeline. Source records, IDs and individual predictions do not enter the app or submission.

Timing and evaluation contracts

Synthetic features use five trailing observations, while the UI waits for nine. Real eligible windows require 24 contiguous current/past observations and eight complete future observations. All current values above 180 mg/dL are excluded. Real static context is age and recorded gender, assumed available at enrollment. All undated labs and outcome-derived summary fields are excluded.

Evidence: public/model-evidence.json, model/real-benchmark-evidence.json. Full protocol: model/README.md and model/REAL_BENCHMARK.md.

Safety boundaries and evidence limits

Team dev / J R Deva Dattan / Army Institute of Technology, Pune

Runtime behavior

Withhold forecasts for disconnected, invalid, stale or insufficient data and for a current above-range reading. The 30-minute stale limit is a prototype policy. The event estimate and persistence trajectory are separate models. Validation-residual bands do not guarantee individual clinical coverage. Review actions remain in the current browser session.

Measured retrospective limitation

Real fused PR-AUC 0.3422 versus dynamic-only 0.3398. Paired 95% patient-bootstrap interval for the difference: -0.00248 to +0.00758. At threshold 0.20, 302 of 326 forecastable episodes are warned with 754 false alerts across about 254 recorded patient-days (2.97/day). The denominator includes gaps and current-high periods. This does not establish meaningful fusion benefit or fitness for care.

Licensing and attribution

Original code: MIT. Dependencies retain their licenses. Real source: Zhao et al., Chinese diabetes datasets for data-driven machine learning, Scientific Data (2023), doi:10.1038/s41597-023-01940-7. Figshare release 5, doi:10.6084/m9.figshare.21600933.v5, CC BY 4.0. Modifications include trailing features and aggregate evaluation. Substantial OpenAI assistance is disclosed in docs/CONTRIBUTIONS.md.